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Coupled Wall Heat Conduction and Coolant Flow Analysis for Liquid Rocket Engines

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Coolant flow modeling in regeneratively cooled rocket engines fed with turbo machinery is a challenging task because of the high wall temperature gradient, the high Reynolds number, the high aspect ratio of the channel cross section, and the heat transfer coupling with the hot-gas flow and the solid material. In this study the effect of wall heat conduction on the coolant flow is analyzed by means of coupled computations between a validated Reynolds-averaged Navier–Stokes equations solver for the coolant flowfield and a Fourier's equation solver for the thermal conduction in the solid material. Computations of supercritical-hydrogen flow in a straight channel with and without coupling with the solid material are performed and compared to understand the role played by the coupling on the coolant flow evolution. Finally, the whole cooling circuit of the space shuttle main engine main combustion chamber is analyzed in detail and discussed for the sake of comparison of results obtained with the present coupled procedure with available literature data.

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Nomenclature

b	=	channel width
c	=	specific heat
D_h	=	hydraulic diameter
E	=	total energy per unit volume
e	=	internal energy
G	=	mass flow rate per unit area
h	=	channel height
h_c	=	coolant side heat transfer coefficient
h_g	=	hot-gas side heat transfer coefficient
k	=	thermal conductivity
l	=	channel length
$\dot{m}$	=	mass flow rate
$\mathbf{n}$	=	unit vector normal to control surface
p	=	pressure
Q	=	heat transfer rate
q	=	heat flux
Re	=	Reynolds number, GD_h/μ_i
r	=	chamber radius
S	=	control surface
s	=	curvilinear abscissa
s_w	=	internal wall thickness
T	=	temperature
t	=	time
t_w	=	rib thickness
u	=	velocity
V	=	control volume
x, y, z	=	Cartesian coordinates
y^+	=	nondimensional wall distance
μ	=	dynamic viscosity
ρ	=	density
τ_{ij}	=	stress tensor

Subscripts

0	=	stagnation
aw	=	adiabatic wall
c	=	combustion chamber
e	=	exit
h	=	hot-gas side
i	=	inlet
w	=	wall

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I. Introduction

TO KEEP the temperature of the thrust chamber walls within their allowed limits despite the huge hot-gas temperature, an intense cooling effort is necessary, which, in case of regenerative cooling, is achieved by flowing one of the propellants into suitable channels surrounding the thrust chamber (Fig. 1). Cooling channels of high-performance rocket engines are generally characterized by high aspect ratio of the cross section (high aspect ratio cooling channels, HARCC) because this feature provides a significant increase in cooling efficiency [1].

The analysis of the flow behavior and the heat transfer characteristics in a regeneratively cooled thrust chamber requires costly experiments or complex numerical simulations, because of the three-dimensional shape of the cooling channels and of the coupling among evolution in hot-gas flow, coolant flow, and wall heat transfer. Although reasonable estimations have been obtained by one-dimensional models with Nusselt-type empirical relations for both coolant and hot-gas flows [2–9], such empirical methods require a wide experimental database to get parameter calibration. To overcome the simplification arising from one-dimensional modeling of hot-gas expansion, efforts have been made to describe the hot-gas flowfield by means of multidimensional computational fluid dynamic (CFD) solvers [10–17], while maintaining semi-empirical modeling for the coolant flow. A suitable calibration of empirical laws with experimental data that are not available in the design phase is, however, still required for the coolant flow, weakening the prediction capability. Therefore, CFD solvers with increasing model complexity have been introduced to study the coolant flow [18–23]. The most relevant shortcoming of these studies is that the wall temperature distribution along the channel periphery is imposed as boundary condition, rather than computed as the result of the heat transfer balance between the hot gas and the coolant through the heat conduction in the solid material that surrounds the channel. As a consequence, the temperature distribution could be far from that enforced as boundary condition, unless coupled CFD analyses of the

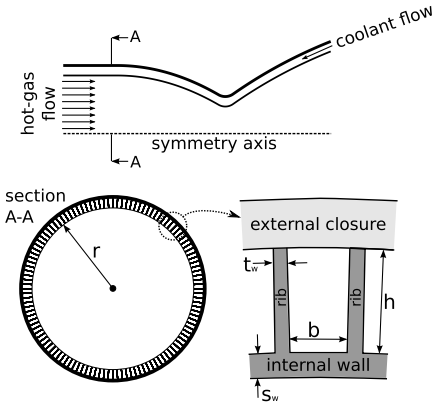

Fig. 1 Schematic of cooling-channels geometry.

coolant flow and the wall structure, eventually considering also the hot-gas flow, are performed [24–27].

The objective of the present study is to present and demonstrate on practical applications a computational procedure able to describe the coupled hot-gas/wall/coolant environment of liquid rocket engines. The coupled solution of the wall heat conduction and the coolant flow is achieved by an iterative procedure in which the wall temperature distribution obtained by the wall heat conduction solver is used as boundary condition for the coolant flow computations. The coolant flow is described by means of a validated CFD solver developed by the authors [21–23,28]. The wall temperature on the lateral wall is computed by means of a numerical solver of the three-dimensional Fourier's problem for the heat conduction in the solid material. Computations of supercritical-hydrogen flow in a straight channel with and without coupling with the solid material are performed and compared to understand the role played by the coupling on the coolant flow evolution. Finally, the whole cooling circuit of the space shuttle main engine (SSME) main combustion chamber (MCC) is analyzed in detail and discussed for the sake of comparison of results obtained with the present coupled procedure with available literature data.

II. Physical and Mathematical Model

The present coupled approach accounts for convection from hot gas, conduction within the wall, and convection to the cooling channels. In fact, regenerative cooling can be modeled as a steady heat flux from a hot gas to a cold fluid through a solid wall. The study of the heat flux between two fluids separated by a solid wall can be divided into three subproblems, which are defined as follows:

1) Hot gas: The propulsive flow of a mixture of gases in a rocket engine, including combustion chamber and converging — diverging nozzle.

2) Wall: The heat conduction through the wall of the rocket engine between the hot gas and the coolant.

3) Coolant: The turbulent flow of the coolant in the channels surrounding the rocket engine.

Hot-gas properties are considered as a datum in the present study. Therefore, the way the hot-gas flow is handled will only be shortly described in Sec. III.

A. Coolant Flow

The adopted integral governing equations of the compressible coolant flow are the Reynolds-averaged Navier–Stokes equations, that is, mass, momentum, and energy equations, which can be written in conservative form for a fixed volume V (bounded by a surface S) in a Cartesian frame of reference, as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \boldsymbol{U}\,\mathrm{d}V + \oint_S \sum_{j=1}^{3} \boldsymbol{F}_j n_j\,\mathrm{d}S = \oint_S \sum_{j=1}^{3} \boldsymbol{G}_j n_j\,\mathrm{d}S \tag{1}$$

In Eq. (1), $\boldsymbol{U}$ is the vector of conserved variables, $\boldsymbol{F}_j$ is the vector of the Eulerian fluxes, and $\boldsymbol{G}_j$ is the vector of viscous fluxes:

$$\boldsymbol{U} = \begin{pmatrix} \rho \\ \rho u_1 \\ \rho u_2 \\ \rho u_3 \\ E \end{pmatrix}, \qquad \boldsymbol{F}_j = \begin{pmatrix} \rho u_j \\ \rho u_1 u_j + p\delta_{1j} \\ \rho u_2 u_j + p\delta_{2j} \\ \rho u_3 u_j + p\delta_{3j} \\ (E+p)u_j \end{pmatrix},$$

$$\boldsymbol{G}_j = \begin{pmatrix} 0 \\ \tau_{1j} \\ \tau_{2j} \\ \tau_{3j} \\ \sum_{i=1}^{3} \tau_{ij} u_i - q_j \end{pmatrix} \qquad (j = 1, 2, 3) \tag{2}$$

where ρ is the density of the fluid, p is the pressure, u_i is the ith component of the velocity, $E = \rho(e + \sum_{i=1}^{3} \frac{1}{2} u_i u_i)$ is the total energy per unit volume, e is the specific internal energy, τ_{ij} is the viscous molecular or turbulent stress tensor, q_j is the jth component of the vector of molecular or turbulent heat flux, considered positive as it is outward-facing the volume V, n_j is the jth component of the unit vector orthogonal to the surface S, considered positive as it is outward-facing the volume V, and δ_{ij} is the Kronecker delta. The molecular part of the stress tensor has been modeled as a general Newtonian fluid and the heat flux by the Fourier's law, while the turbulent part of stress tensor and heat flux vector have been modeled according to the one-equation Spalart — Allmaras turbulence model [29]. This turbulence model, which is able to describe the near-wall turbulence, has been successfully used for a multitude of flow problems [30], including the cooling-channel flow problem [22]. The fluid considered in the present study is para-hydrogen, which is the hydrogen modification of interest for liquid rocket engines [31]. Para-hydrogen is described by the modified Benedict — Webb — Rubin equation of state [32] for temperatures below 400 K and by the calorically and thermically perfect gas model for temperatures above 400 K, where the former model is no longer valid. The fluid viscosity and thermal conductivity below 400 K are described by proper relations taken from [32], and by the Sutherland's model above 400 K [33]. The models used below and above the threshold temperature of 400 K are matched such that the equation of state and the molecular properties do not present discontinuities.

B. Wall Conduction

In case of wall material heat conduction, the integral governing equation is given by the principle of energy conservation and the Fourier's law, which can be written in conservative form for a fixed volume V (bounded by a surface S) in a Cartesian frame of reference, as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho_w c_w T_w\,\mathrm{d}V = -\oint_S \sum_{j=1}^{3} q_{w,j} n_j\,\mathrm{d}S \tag{3}$$

In Eq. (2), ρ_w, c_w, and T_w are the material density, specific heat, and temperature, respectively, and $q_{w,j}$, is the jth component of the conductive heat flux, considered positive as it is outward-facing the volume V, and that, according to the Fourier's law, is expressed as:

$$q_{w,j} = -k_w \frac{\partial T_w}{\partial x_j} \tag{4}$$

where k_w is the material thermal conductivity.

III. Numerical Method

Numerical discretization and solution of the mathematical model is achieved by means of finite volume approaches and a suitable coupling technique to reach steady-state solution of the fluid and thermal behavior in the regenerative cooling system. Details of the numerical approaches considered for the coolant flow and wall

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conduction computations, together with a description of the approach followed to couple the two solutions, are given here in the following.

A. Coolant Flow

The numerical solution of the governing equations is carried out by a time-dependent gasdynamics solver, which is based on a Godunov-type finite volume method [21,34]. The fluid domain is divided into control volumes and, for the generic hexahedral control volume, the governing equations are written as:

$$\left(\frac{\mathrm{d}\bar{\boldsymbol{U}}}{\mathrm{d}t}\right)V + \sum_{k=1}^{6}\sum_{j=1}^{3}\tilde{\boldsymbol{F}}_j^k n_j^k S^k = \sum_{k=1}^{6}\sum_{j=1}^{3}\tilde{\boldsymbol{G}}_j^k n_j^k S^k \tag{5}$$

where the k summation is over all the six faces of the control volume. The overbar indicates the average value of the conserved variables in the control volume, while the tilde indicates the numerical flux (Eulerian and viscous) through face S^k. To obtain second-order accuracy in space, the viscous fluxes are evaluated by second-order-accurate central differencing while the Eulerian fluxes are evaluated by the solution of Riemann problem at the cell interfaces and by linear cell reconstruction of primitive variables, using a minmod slope limiter to satisfy the total variation diminshing stability condition. Steady-state solution is reached by carrying out time integration until the mass equation residual, which is based on flowfield density variation between two consecutive time steps, is reduced by a reasonable quantity (6 to 8 orders of magnitude generally suffices to reach steady state). To speed up convergence to steady state, local time step is used. The main feature of the numerical solver is that the governing equations have been considered in the conservation form and the fluid thermodynamic properties are stored in a look-up table [21]. The database is based on ρ and p as independent variables and on regular discretization Δp and $\Delta\rho$. The requested dependent variables are computed by bilinear interpolation. This approach permits to get the desired value without any iterative searching algorithm [21]. Validation of the present numerical approach has been reported by comparison made with available experimental data on supercritical-pressure nitrogen [21,22] and hydrogen [28].

B. Wall Conduction

Similarly to the coolant flow problem, the numerical solution of the governing equation [Eq. (3)] for the wall conduction is carried out by a time-dependent solver that is based on a finite volume method. Thus, the solid domain is divided into control volumes and, for the generic hexahedral control volume, the governing equation is written as:

$$\left(\frac{\mathrm{d}\bar{T}_w}{\mathrm{d}t}\right)V = \frac{k_w}{\rho_w c_w}\sum_{k=1}^{6}\sum_{j=1}^{3}\left(\widetilde{\frac{\partial T_w}{\partial x_j}}\right)^k n_j^k S^k \tag{6}$$

where the k summation is over all the six faces of the control volume and the solid properties (ρ_w, c_w, and k_w) are considered constant. The overbar indicates the average value of the temperature in the control volume while the tilde indicates the numerical temperature gradient in the face S^k. To obtain second-order accuracy in space, the temperature gradients are evaluated by second-order-accurate central differencing. Steady-state solution is reached by carrying out time integration until the equation residual, which is based on temperature variation between two consecutive time steps, is reduced by a reasonable quantity (6 to 8 orders of magnitude generally suffices to reach steady state). To speed up convergence to steady state, local time step can be used.

C. Coupling

The subproblems are coupled by the two steady-state balances of three heat fluxes: from hot gas to wall; through the wall; and from wall to coolant. The thermal hot-gas behavior is described by means of the heat transfer coefficient h_g and the adiabatic wall temperature T_{aw}. In this way the hot-gas problem is decoupled from the wall and coolant problem because the heat transfer coefficient weakly depends on wall temperature [17] and the adiabatic wall temperature is representative of the total energy content of the hot gas. Of course, neither hot-gas wall temperature nor heat flux are known, because they depend on the thermal equilibrium between flows through wall by the relation:

$$q_{w,h} = h_g(T_{\mathrm{aw}} - T_{w,h}) \tag{7}$$

In the present study the hot-gas thermal characterization is a known input in terms of h_g and T_{aw}. With these boundary conditions, the coupled solution of the wall heat conduction and the coolant flow is achieved by an iterative procedure, in which the distribution of the coolant-side wall temperature obtained with the Fourier's equation solver is used as a boundary condition for the coolant flow computations. In particular, the coupling procedure is described by the following steps:

0. RANS computation of the coolant flow imposing a first-tentative temperature distribution of the coolant-side wall as boundary condition and evaluation of the heat transfer coefficient of the coolant-side wall, h_c, from the computed heat flux, the enforced wall temperature, and a reference coolant temperature, set equal to the inlet coolant temperature T_i.

1. Wall conduction computation imposing the distributions of the heat transfer coefficient and reference temperature as boundary conditions: h_g and T_{aw} on the hot-gas side and h_c and T_i on the coolant side. This analysis provides a resulting distribution of T_w in the solid material.

2. RANS computation of the coolant flow imposing the wall temperature provided in step 1 as boundary condition and evaluation of the coolant heat transfer coefficient along the channel perimeter, h_c, from the computed heat flux, the enforced wall temperature, and the reference coolant temperature T_i.

Iterations are made repeating steps 1 and 2. The process ends when the difference between the computed values of $T_{w,h}$ in two consecutive iterations is smaller than an assigned tolerance. In the present study three/four iterations have shown to be sufficient to achieve a prescribed tolerance of 20 K.

IV. Straight Channel

A. Test Cases

The coupled analysis of coolant flow and wall material heat transfer is performed for a straight channel test case. The geometric features are: channel base $b = 2$ mm, channel height $h = 16$ mm (that is, aspect ratio 8), rib base $t_w = 1$ mm, internal wall thickness $s_w = 1$ mm and channel length $l = 400$ mm, which corresponds to 112.5 hydraulic diameters. Dimensions of the cooling-channel cross-section are similar to those of the Vulcain engine thrust chamber [19] and the relatively long channel ensures that the flow is affected by the inlet configuration only by a small fraction of the channel length. The imposed inlet conditions are $T_i = 80$ K and $\dot{m} = 134.4$ g/s while the outflow static pressure is $p_e = 60$ bar. The hot-gas side boundary conditions are $h_g = 13\ \mathrm{kW/m^2\,K}$ and $T_{\mathrm{aw}} = 3500$ K; adiabatic conditions are considered at the external-wall interface for both the wall and the coolant. Hot-gas side characterization is representative of the Vulcain engine combustion chamber [19], where the wall temperature is between 700 and 800 K and the average wall heat flux is $\sim$36 $\mathrm{MW/m^2}$ (and thus, from Eq. (7), $h_g \sim 13\ \mathrm{kW/m^2\,K}$). The considered coolant flow is supercritical hydrogen while the solid material properties refer to high-conductivity copper alloy [35], which is widely used in regeneratively-cooled rocket engines with rectangular cooling channels. In particular, the material properties are considered constant with temperature: $\rho_w = 8920\ \mathrm{kg/m^3}$, $c_w = 380\ \mathrm{J/kg\,K}$, and $k_w = 390\ \mathrm{W/m\,K}$. The flow conditions of the presented test cases reproduce many features of the actual rocket engine cooling passages: high-Re turbulent flow ($Re = 3.2 \cdot 10^6$ at the inlet section of the present test case), high aspect ratio of the cross section, wall-coolant coupling that induces radial thermal stratification, large entering heat flux, highly-compressed state of the coolant (due to the low-temperature and supercritical-pressure inlet condition).

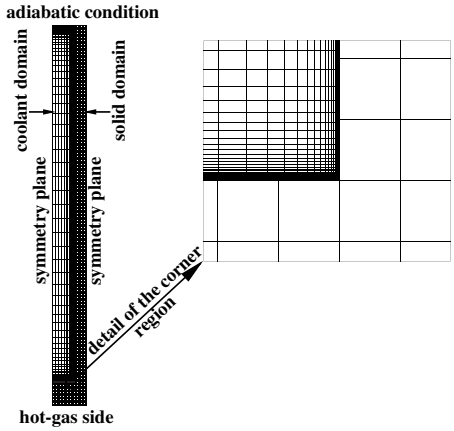


Fig. 2 Computational meshes and boundary conditions of the straight-channel test case.

To understand the role of the solid material heat conduction on the coolant flow, the coupled solution is compared to that without wall material heat conduction (uncoupled solution). The uncoupled test case is achieved by means of the CFD solver, for the same channel geometry and inlet/exit boundary conditions of the coupled test case. The imposed wall temperature at the channel bases are $T_w = 80$ K at the cold side and $T_w = 600$ K at the hot side. A linear distribution is considered on the lateral wall to roughly simulate the cooling-channel fin effect. Note that the wall temperature at the cold side is equal to the coolant inlet temperature to approximate the adiabatic-wall condition of the coupled test case and the wall temperature at the hot-side is the average temperature of the wall temperature at the hot side evaluated for the coupled solution.

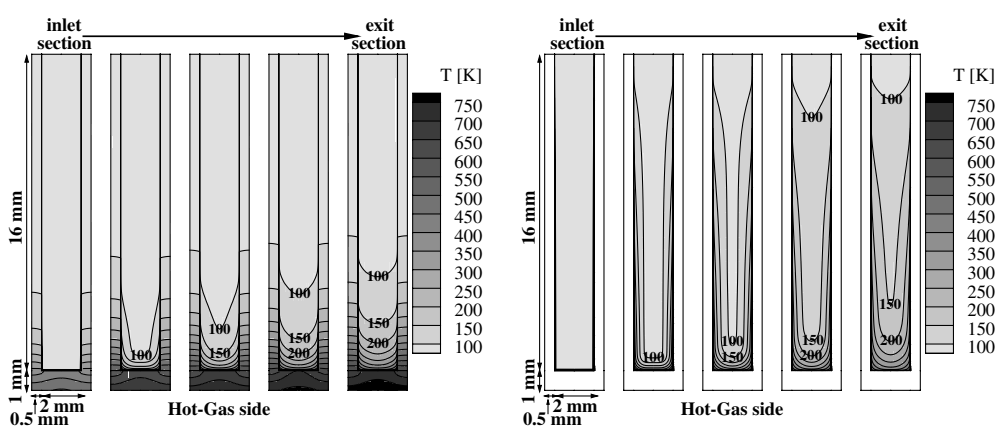
The reference computational grid of the coolant domain is composed of $32 \times 128 \times 64 \simeq 2.6 \cdot 10^5$ volumes along the half-base, the height and the axial direction of the channel, respectively. Taking advantage from the flow symmetry with respect to the vertical mid-plane, only one-half of the physical domain is discretized. The volumes are clustered near the walls to accurately describe the turbulent boundary layers (the non-dimensional distance y^+ at the wall-adjacent cell is of order 1) while the axial distribution is uniform. The computational grid of the solid domain is composed of $2432 \times 64 \simeq 1.6 \cdot 10^5$ volumes along the cross section and the axial direction of the channel, respectively. Each volume has a squared cross section with a side length of 62.5 μm. Also in this case only one-half of the physical domain is discretized. Computational mesh of the channel cross section is presented in Fig. 2. To verify that the selected grids with their mesh distributions are suitable to accurately describe the present test case, grid convergence analyses have been performed for the uncoupled coolant flow and the uncoupled thermal problem in the solid domain. The analysis of the coolant-flow mesh has been presented in [22] and that of the solid-domain has shown that a further increase of the mesh discretization does not change appreciably the solution.

Note that, because of the mesh stretching required at wall to describe the fluid boundary layer, the coolant-domain volumes that face the coolant-wall interface are far smaller than those of the solid-domain, especially in the corner region (Fig. 2). As a consequence, the coolant-side heat transfer coefficient to be used as boundary condition of the wall-conduction solver is evaluated considering the average heat flux computed by the coolant flow solver and that passes through the boundary faces of the solid-domain. On the other side, the wall temperature computed by the wall-conduction solver is linearly interpolated on the finer boundary of the coolant flow solver mesh.

B. Results

The thermal behavior of straight HARCC is shown by temperature fields of both coolant and wall material in Fig. 3, where the evolution of temperature is shown at different cross-sections, from the inlet toward the exit. Cross sections are located every 100 mm along the channel stream-wise direction. In Fig. 3b the wall material is shown in white to emphasize that it is neglected in the uncoupled computation. In both computations, with and without accounting for wall conduction, due to the low-temperature and high-pressure inlet condition ($T_i = 80$ K and supercritical pressure), hydrogen enters the channel in a highly-compressed state, its density being nearly 24 $\mathrm{kg/m^3}$. Then, as it is heated by the entering heat flux from the hot walls located in the lower part of the channel section, the fluid accelerates, while its pressure decreases, due to the friction force acting on the channel wall and the entering heat flux. The pressure distribution (not shown here) is one-dimensional like, regardless of the three-dimensional channel geometry and the asymmetric thermal boundary conditions. The one-dimensional behavior of pressure is a direct consequence of the constant cross-section, straight channel geometry. Figure 3 shows that the high temperature flow region grows as the fluid moves from the inlet to the exit section, but it remains close to the hot wall along the whole length, for both the coupled and the uncoupled computations. As a consequence, the fluid far from the bottom wall is nearly at the inlet condition and an evident non-uniform thermal distribution occurs.

The wall temperature increases along the channel length, its maximum value being 506 K at the channel inlet and 793 K at the channel exit (Fig. 3a), whereas it is imposed and constant in the uncoupled analysis. The maximum value of the wall temperature is located on the hot-gas side and on coolant symmetry-plane, that is, at the maximum distance from the wall ribs because coolant flow thermal resistance is larger than that of the copper wall ribs. The lower thermal resistance makes the wall ribs a preferred path for the heat flux and thus a temperature gradient is present also in the transversal direction (i.e., base-direction): at the hot-gas side of the exit section the wall temperature variation is 20.9 K. Comparison evidences that considering the material heat conduction leads to a temperature



a) Coupled computation **b) Uncoupled computation**

Fig. 3 Temperature fields.

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distribution on the side wall which is much less uniform than the linear distribution imposed in the uncoupled computation. In particular, in the upper part of the channel the wall temperature is nearly constant and its value is that of hydrogen at the inlet section (i.e., $T_i = 80$ K). This means that the upper part of the channel does not provide heat to the coolant and thus the total amount of the heat provided to the coolant is lower if the wall conduction is taken into account. By means of the one-dimensional energy balance, the heat transfer rate that enters the channel Q is proportional to the increase of the coolant stagnation enthalpy, resulting in a larger coolant stagnation temperature gain with a larger heat transfer rate. Stagnation temperature increases 72.3 K in case of linear wall temperature and 20.6 K in case of coupled analysis. Obviously, the lower amount of entering heat also leads to lower coolant pressure drop between the channel inlet and exit sections: 10.2 bar in case of linear wall temperature and 6.3 bar in the coupled case. The large difference between the two models is due to the high aspect ratio which leaves a large portion of the coolant mass flow rate not affected by the heat flux from hot-gas and thus yields a non-uniform thermal stratification in the solid material (Fig. 3a).

V. Space Shuttle Main Engine

A. Test Cases

For the sake of comparison with experimental data, a coupled coolant/wall analysis is also carried out for the SSME MCC, whose data relevant to the regenerative cooling circuit have been published in the open literature [10,36,37]. The SSME is a LOX/hydrogen staged-combustion engine that uses the fuel as coolant and presents a throttle range between 67 and 109% of rated thrust. In particular, the selected data refer to the standard throat version of the thrust chamber, which consists in a coolant liner made of NARloy-Z copper alloy and has a throat diameter of 261.75 mm. The liner provides the coolant flowpath for the MCC. It has 390 milled axial coolant channels that are closed out by an electroforming process that deposits a copper barrier, followed by a nickel Inconel 718 structural closeout over the coolant channels [10,36]. The channel geometry properties, taken from [36], are reported in Fig. 4. Note that, close to the chamber injector plate, located 355.6 mm upstream of the throat, the channel height increases by almost three times to reduce the coolant velocity at the inlet of the exit manifold. However, to prevent numerical issues at the exit boundary conditions due to the large area variation of the channel cross section, the channel height has been recasted as illustrated in Fig. 4.

Among the available data, the most complete set for the standard throat SSME-MCC refer to the full power level (FPL) operative condition (109% of rated thrust) [10,36], with chamber pressure $p_c = 225.87$ bar and oxidizer-to-fuel mixture ratio $O/F = 6.0$. The coolant operative conditions at FPL are taken from [10]: overall coolant mass flow rate is $\dot{m} = 14.306$ kg/s (i.e., 36.68 g/s for a single channel), inlet temperature is 53.89 K and inlet pressure is 445.47 bar. Moreover, on the basis of the one-dimensional coolant flow computations performed in [10], the exit temperature is 255.56 K and the exit pressure is 354.18 bar. These data are used to enforce the following coolant hydrogen inlet and exit conditions for FPL computation: mass flow rate, inlet temperature and exit pressure. Coolant thermodynamic and transport properties used in the fluid solver are taken from [32], while the wall material properties are taken from [35] and refer to those of NARloy-Z copper alloy at 533 K: $\rho_w = 9032$ kg/m^3, $c_w = 397$ J/kg K, $k_w = 316$ W/m K. Unfortunately, the available data do not include the whole information needed for the present simulations. In fact, besides geometrical data, inlet, and outlet flow conditions, the hot-gas-side adiabatic wall temperature and heat transfer coefficient also have to be provided as a boundary condition for the coupled approach presented in Sec. III.C. To get these lacking data, some useful information has been found in [37], where the hot-gas side heat transfer coefficient is provided at the rated power level (RPL) operative condition (100% of rated thrust, with chamber pressure $p_c = 204.7$ bar and oxidizer-to-fuel mixture ratio $O/F = 6.0$). With RPL data, a reasonable estimation of the hot-gas heat transfer coefficient of the FPL condition can be obtained by means of the same scaling technique as that presented in [37]. In particular, the adopted scaling technique is based on the classic relationship of $h_g \propto p_c^{0.8}$. However, as suggested in [37], from the injector face to approximately 127 mm downstream, the heat transfer coefficient of the SSME/MCC is imposed equal to that of RPL data because the heat transfer rates near the injector were primarily influenced by the distance from the injector. As to adiabatic wall temperature of the hot gas, it is evaluated by means of the recovery factor and using one-dimensional chemical equilibrium [38] at the appropriate test chamber pressure and mixture ratio, as also described in [37]. The RPL heat transfer coefficient data, its scaled value for the FPL operative conditions, and the computed adiabatic-wall temperature profiles are shown in Fig. 5. In particular, Fig. 5 shows the values of h_g and T_{aw} assumed for the FPL condition that are higher than the corresponding RPL values due to the higher chamber pressure.

The computational grid of the coolant domain is composed of $64 \times 192 \times 256$ volumes along the half-base, the height, and the axial direction of the channel, respectively. As for the straight channel test case (Sec. IV.A), because of the flow symmetry with respect to the vertical midplane, only one-half of the physical domain is discretized. The volumes are clustered near the walls to accurately describe the turbulent boundary layers (the nondimensional distance y^+ at the wall-adjacent cell is of order 1) while the axial distribution is uniform. The computational grid of the solid domain is divided by 256 volumes in the axial direction and by 1536 rectangular-volumes in the cross section. Also in this case only one-half of the physical domain is discretized. Computational mesh of the axial view and of various the channel cross sections is presented in Fig. 6. Some volumes on the mesh are excluded in the figure to aid in viewing.

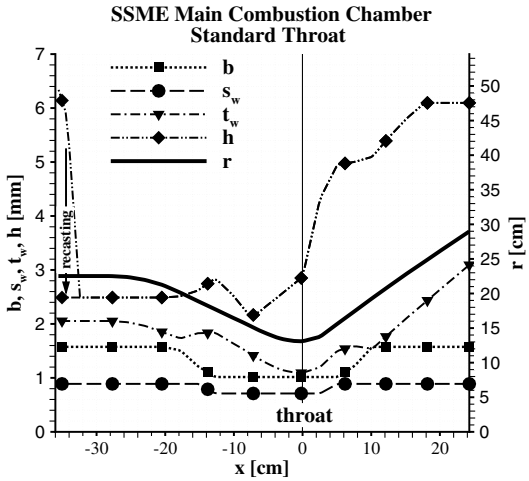


Fig. 4 SSME main combustion chamber geometry (data taken from [36]).

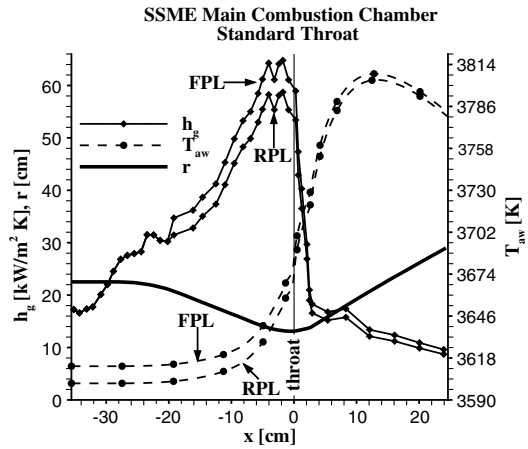


Fig. 5 SSME MCC hot-gas heat transfer coefficient and adiabatic wall temperature profiles for the RPL and FPL condition.

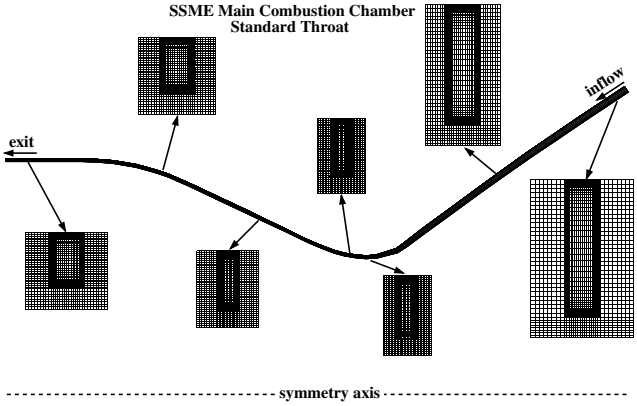
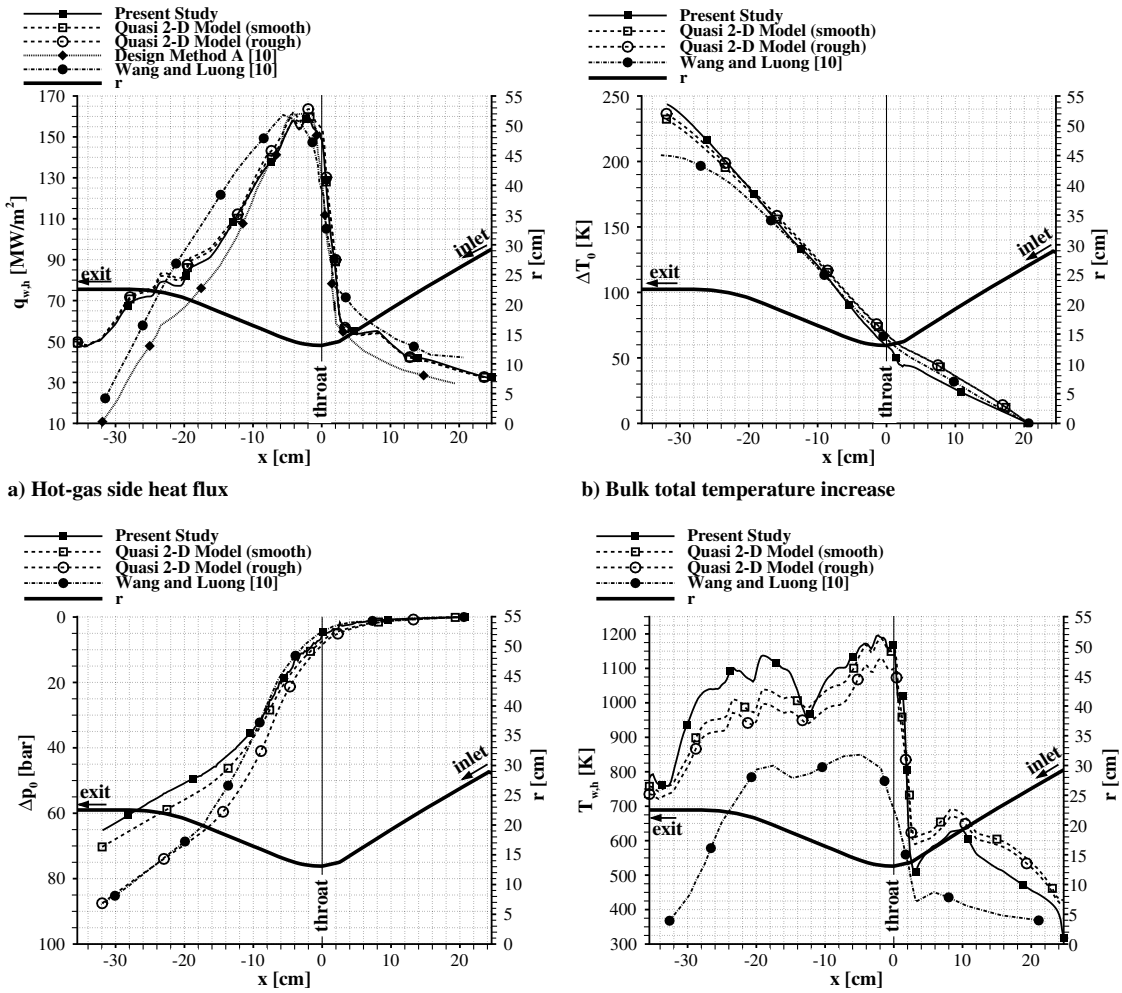


Fig. 6 Computational meshes of the SSME test case.

B. Results

The results obtained with the present approach are compared with available literature data in Fig. 7. Besides data reported in [10], which include the heat fluxes predicted by Rocketdyne (referred to as " Design Method A"), also results obtained with the "quasi-2-D" model developed by the authors [9] are shown for the sake of completeness and to ease discussion. The quasi-2-D model relies on a basically one-dimensional approach, with empirical correlations for the evaluation of wall heat fluxes and friction, over which a correction to take into account coolant and wall thermal stratification in the radial direction has been developed, implemented, and validated. In particular, results relevant to the quasi-2-D model are obtained with the same boundary conditions as for the present 3-D-coupled simulation and two different levels of wall roughness: 0 μm (smooth) and 0.18 μm (rough). The comparison of hot-gas side heat flux shows a good agreement among all results (Fig. 7a). Note that for the present 3-D approach the plotted distributions are those on the cooling-channel symmetry plane, being however the variations in the base-direction small compared with axial variations and thus negligible in this plot.

The converged solution obtained with the present numerical approach shows a maximum heat flux of 159.7 $\mathrm{MW/m^2}$ at the throat, which corresponds to the maximum value of heat transfer coefficient (Fig. 5). The present peak value is very close to those estimated in [10] and reported in Fig. 7a: 161.5 $\mathrm{MW/m^2}$ (Design Method A) and 160.8 $\mathrm{MW/m^2}$ (Wang and Luong [10]). Moreover, the computed result is within those presented in [10] all along the chamber but in the barrel section of the engine ($x < -23$ cm) where the hot-gas side heat transfer is strongly influenced by the injector-elements [37]. The quasi-2-D model computation shows in both smooth and rough cases a behavior quite close to that of the present 3-D solution. The coolant temperature increase is shown in Fig. 7b, where it can be seen that differences among curves are always within 15 K, except at the final part of cooling circuit, where most of discrepancy builds up, with a final value of about 40 K. However, this is explained by the different behavior of heat flux in the barrel section of the engine (Fig. 7a). Three-dimensional effects can be remarked by comparison of the 3-D and the quasi-2-D simulations; evident effects are present just at the right of the throat, where the coolant bulk temperature changes its growth gradient and in the final part of the cooling circuit. In these



a) Hot-gas side heat flux

b) Bulk total temperature increase

c) Total pressure loss

d) Hot-gas side wall temperature

Fig. 7 SSME main combustion chamber, standard throat, FPL operative condition. Comparison of present and literature numerical simulations along the engine axis.

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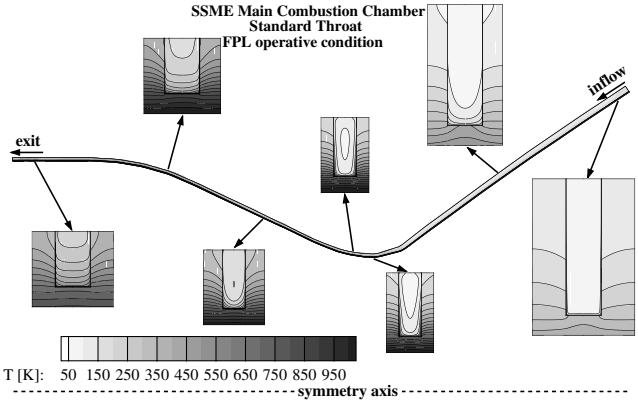

Fig. 8 Coolant and wall temperature evolution.

regions the two estimations differ by nearly 15 K. Pressure loss behavior is presented in Fig. 7c. The pressure loss estimated with the 3-D approach is 65.2 bar, which is 25% lower than that of [10] (87.9 bar). The pressure loss discrepancy could be partially attributed to the relevant wall temperature differences between the literature solution and the 3-D simulation that will be shown in the following and yields to different coolant friction. However, the pressure loss discrepancy could also be explained by the assumption of smooth walls. In fact, the quasi-2-D model results fairly agree with present simulation if the wall is assumed smooth (i.e., difference of 7.8%) whereas it agrees with results of [10] if the mentioned level of roughness is assumed.

Differently from other variables, larger differences can be seen analyzing the wall temperature behavior. In particular, the computed hot-gas side wall temperature is much larger than expected, the maximum value in the throat region being 1196 K for the present model and nearly 870 K as declared in the open literature [10,35]. However, the qualitative behavior of $T_{w,h}$ is similar to that reported in [10]: a peak value slightly upstream of the throat, a local minimum at nearly 130 mm upstream of the throat, and a strong variation downstream of the throat, followed by a local maximum. The agreement with the quasi-2-D solutions obtained with the same input data indicates that the large differences with respect to [10] are due to different heat transfer coefficient and/or adiabatic wall temperature values. In fact, according to Fig. 5, the maximum hot-gas side wall temperature of the FPL condition can be evaluated considering the maximum heat transfer coefficient, located at 17.7 mm upstream the throat and equal to $h_{h,\max} = 64.8\ \mathrm{kW/m^2\,K}$, the corresponding adiabatic-wall temperature, $T_{\mathrm{aw}} = 3664$ K, and the nominal heat flux in the throat region, which is expected to be $160\ \mathrm{MW/m^2}$ [10,26]. By means of Eq. (7), the resulting maximum wall temperature is 1195 K, which is very close to those obtained by the present numerical model. As a consequence, the hot-gas side heat transfer coefficient and/or adiabatic wall temperature taken from [37] result too high with respect to those of [10]. The larger values may be attributed to the limits of validity of the scaling laws, which are considered in the estimation of the heat transfer coefficient and the adiabatic wall temperature [37]. Analyzing the wall temperature behavior, it can also be made evident that wall temperature predicted by the 3-D approach is slightly larger than that predicted by the quasi-2-D model upstream of the throat section. This difference is of the order of 100 K if the smooth case is considered. It is also worth noticing that the beneficial effect of the wall roughness on the wall temperature (estimated with the quasi-2-D model) implies a reduction of nearly 60 K. Finally, comparing the 3-D and the quasi-2-D simulations, evident differences can be noted in the barrel section and in the convergent part of the engine: the wall temperature evaluated with the 3-D computation shows a sudden drop at $x \sim -10$ cm and it is 100 K larger in the region between $x \sim -30$ cm and $x \sim -15\ cm$. These peculiar behaviors are only captured by the 3-D coupled simulation.

The multidimensional coolant and wall temperature evolution at various cross sections of the SSME main combustion chamber cooling circuit and for the FPL operative condition is illustrated in Fig. 8. The thermal stratification that occurs in the radial direction for both the coolant and the wall material can be easily noticed; in particular, due to the relatively high thermal conductivity of the copper alloy, the thermal stratification in the solid ribs develops starting from the first channel sections and results in a rather uniform wall temperature gradient in the radial direction. These results show that, as the SSME MCC is not characterized by channels with high aspect ratio of the cross section, the nonuniform wall temperature distribution noticed in Sec. IV.B is not present here and, thus, the hypothesis of linear wall temperature could have been adopted [39]. Moreover, the coolant temperature gradient is mostly located in the inner part of the channel section and thus, far from the inner base, the coolant temperature is rather uniform. Only in the final part of the cooling circuit, namely in the barrel section of the combustion chamber, the coolant presents a uniform thermal stratification in the radial direction.

VI. Conclusions

A loosely coupled procedure for the simulation of the behavior of wall and coolant flow in a regeneratively cooled liquid rocket engine has been set up and demonstrated on two different test cases: a straight asymmetrically heated channel with high aspect ratio of the cross section and a regenerative cooling circuit of an operative rocket engine. The numerical approach is based on two finite volume three-dimensional solvers: a validated Reynolds-averaged Navier–Stokes equations solver for the coolant flow and a Fourier problem solver for the heat conduction in the solid material that surrounds the channel. Results for supercritical hydrogen in heated straight channels with high-aspect-ratio cross section have emphasized that the wall temperature distribution on the side wall is far from linear, especially because of the high aspect ratio of the cross section. The analysis of the whole cooling circuit of the space shuttle main engine main combustion chamber has shown a reasonable agreement with published data as far as wall heat flux distribution and coolant pressure loss are concerned. On the other hand a large discrepancy has been found on wall temperature computation, which may be due to the different hot-gas characterization (heat transfer coefficient and/or adiabatic-wall temperature) with respect to the literature solution. Results have emphasized that, as the channel aspect ratio of this application is rather low in this application, the radial temperature gradient on the lateral wall could also be considered constant. It is therefore in the case of HARCC that a proper wall conduction description is necessary to evaluate the detailed coolant flowfield and heat transfer within a typical liquid rocket engine environment. Finally, the comparison of three-dimensional and quasi-2-D results has shown that significant 3-D effects may occur in some parts of the cooling circuit, which cannot be neglected in a detailed prediction of design parameters as for instance wall temperature.

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